Lab Sheet-02

Object Oriented Programming Using Java

MCA III Semester

Arithmetic Operators (Q1–Q3)

1. Write a Java program to find the area of a triangle given its three sides using **Heron's formula**.
2. Create a Java program to calculate the compound interest given principal, rate, and time.
3. Develop a Java program to find the distance between two points in a 2D plane using the distance formula.

Unary Operators (Q4–Q5)

4. Write a Java program to demonstrate prefix and postfix increment by simulating a counter for visitors entering and leaving a store.
5. Create a Java program that takes an integer and outputs its negative value using unary operators.

Assignment Operator (Q6–Q7)

6. Write a Java program to gradually reduce a number by half using /= until it becomes less than 1, counting the number of steps.
7. Create a Java program to accumulate rainfall data for 7 days using += and find the total.

Relational Operators (Q8–Q9)

8. Write a Java program to check whether three given angles form a valid triangle.
9. Create a Java program to compare two strings for lexicographic order without using built-in compare functions (use relational operators on characters).

Logical Operators (Q10–Q11)

10. Write a Java program to determine if a given year is both a leap year and within a given range.
11. Create a Java program to check if a student passes a course, where they must have at least 40% in theory **AND** 50% in practical **OR** overall, 50%.

Ternary Operator (Q12–Q13)

12. Write a Java program to find the smallest of four numbers using nested ternary operators.
13. Create a Java program to check if a character is a vowel, consonant, digit, or special symbol using the ternary operator.

Bitwise Operators (Q14–Q15)

14. Write a Java program to swap two integers without using a temporary variable (use XOR).

15. Create a Java program to count the number of set bits (1s) in the binary representation of a number using bitwise operators.

Shift Operators (Q16–Q17)

16. Write a Java program to perform fast multiplication and division of a number by powers of two using shift operators.
17. Create a Java program to cyclically rotate the bits of an integer to the left by 2 positions.

B. Loops & Control Statements

for loop (Q18–Q19)

18. Write a Java program to print the first 20 terms of the Fibonacci sequence using a for loop.
19. Create a Java program to print all Armstrong numbers between 1 and 1000 using a for loop.

do-while loop (Q20–Q21)

20. Write a Java program to repeatedly accept a password until the correct one is entered (use do-while).
21. Create a Java program to display all factors of a number using a do-while loop.

while loop (Q22–Q23)

22. Write a Java program to reverse the digits of an integer using a while loop.
23. Create a Java program to determine if a number is a palindrome using a while loop.

for-each loop (Q24–Q25)

24. Write a Java program to find the maximum and minimum values in an integer array using a for-each loop.
25. Create a Java program to calculate the average marks of students stored in a 2D array using a for-each loop.

C. switch-case, continue, break

switch-case (Q26–Q27)

26. Write a Java program that takes a day number (1–7) and displays whether it's a weekday or weekend using switch-case.
27. Create a Java program that takes a grade (A–F) and prints its qualitative meaning (e.g., A → Excellent, B → Good, etc.) using switch-case.

continue statement (Q28)

28. Write a Java program to print numbers from 1 to 50, skipping all numbers that are perfect squares using continue.

break statement (Q29)

29. Create a Java program to generate random integers between 1 and 100 and stop when a number divisible by both 7 and 13 is found using break.

Mixed Advanced (Q30)

30. Write a Java program that:

- Uses **shift operators** to check if a number is a power of 4,
- Uses **bitwise** to toggle the 3rd bit of that number,
- Uses a **for loop** to print its multiplication table but **skips** multiples of 6 using continue,
- Stops if it reaches a multiple of 48 using break.